Practical Digital Filtering and Oversampling

Copyright © Barry Van Veen 2014

AllSignalProcessing.com

Key Concepts

1) Replacing an analog, or continuous-time, filter with a discrete-time filter involves sampling, discrete-time filtering, followed by reconstruction.

    a) Practical sampling requires an analog anti-aliasing filter to be applied prior to sampling.

    b) Practical reconstruction requires a zero-order hold followed by an analog anti-imaging filter.

    c) If aliasing and imaging are prevented, the equivalent analog filtering characteristic within the Nyquist band ($|\Omega| <$

$\Omega_s/2$) is the product of anti-aliasing, anti-imaging, zero-order hold, and discrete-time filter frequency responses.

2) Building analog anti-aliasing and anti-imaging filters with narrow transition bands is difficult and expensive.

3) Oversampling, or increasing the sampling rate beyond that required by the sampling theorem, reduces the cost of the analog filters at the expense of additional computation.

    a) The width of the transition bands in the anti-imaging and anti-aliasing filters is increased, allowing simpler designs.

b) The number of samples per second that must be processed by the system increases, increasing computation and storage cost.

4) One approach to balancing analog filter complexity against processing requirements is to change the sampling rate in discrete-time.

a) The signal is sampled at a high rate relative to the sampling theorem requirements to reduce the cost of the anti-aliasing filter.

b) Computationally efficient discrete-time techniques are used to downsample, or reduce the sampling rate, prior to storage or discrete-time processing.

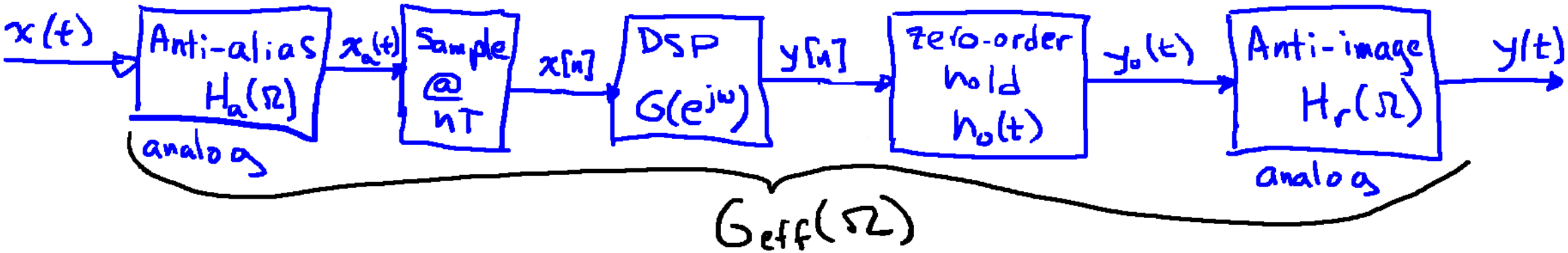
c) After storage or discrete-time processing, computationally efficient discrete-time techniques are used to upsample, or increase the sampling rate, prior to reconstruction.

d) Zero-order hold reconstruction is performed at a high rate relative to the sampling theorem requirements to reduce the cost of the anti-imaging filter.

5) Oversampling approaches trade reduced cost of the analog filters against slightly higher computation requirements and faster sampling and zero-order hold hardware.

Video Screenshots

Practical Digital Filtering and Oversampling



Provided $|H_a(\Omega)| \approx 0 \quad |\Omega| > \frac{\Omega_s}{2}$

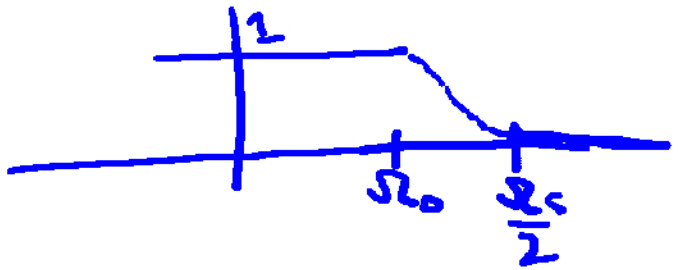
$|H_r(\Omega)| \approx 0 \quad |\Omega| > \frac{\Omega_s}{2}$

$\Rightarrow G_{eff}(\Omega) = \begin{cases} H_a(\Omega) G(e^{j\Omega T}) H_0(\Omega) H_r(\Omega) \\ 0, \quad |\Omega| > \Omega_s/2 \end{cases}$

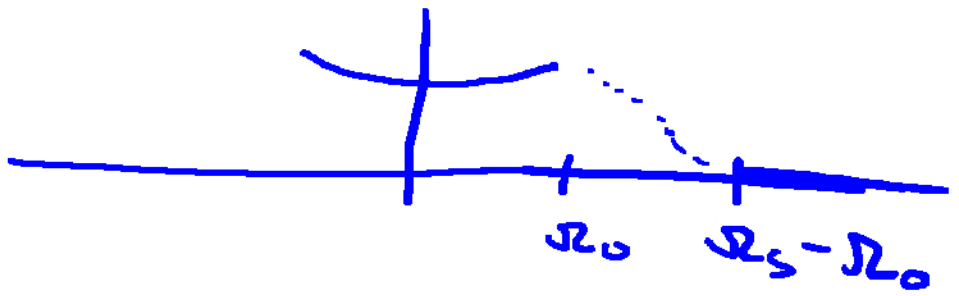
$|\Omega| < \frac{\Omega_s}{2}$

Building narrow transition band filters $H_a(\Omega)$ and $H_r(\Omega)$ is difficult / expensive

$|H_a(\Omega)|$



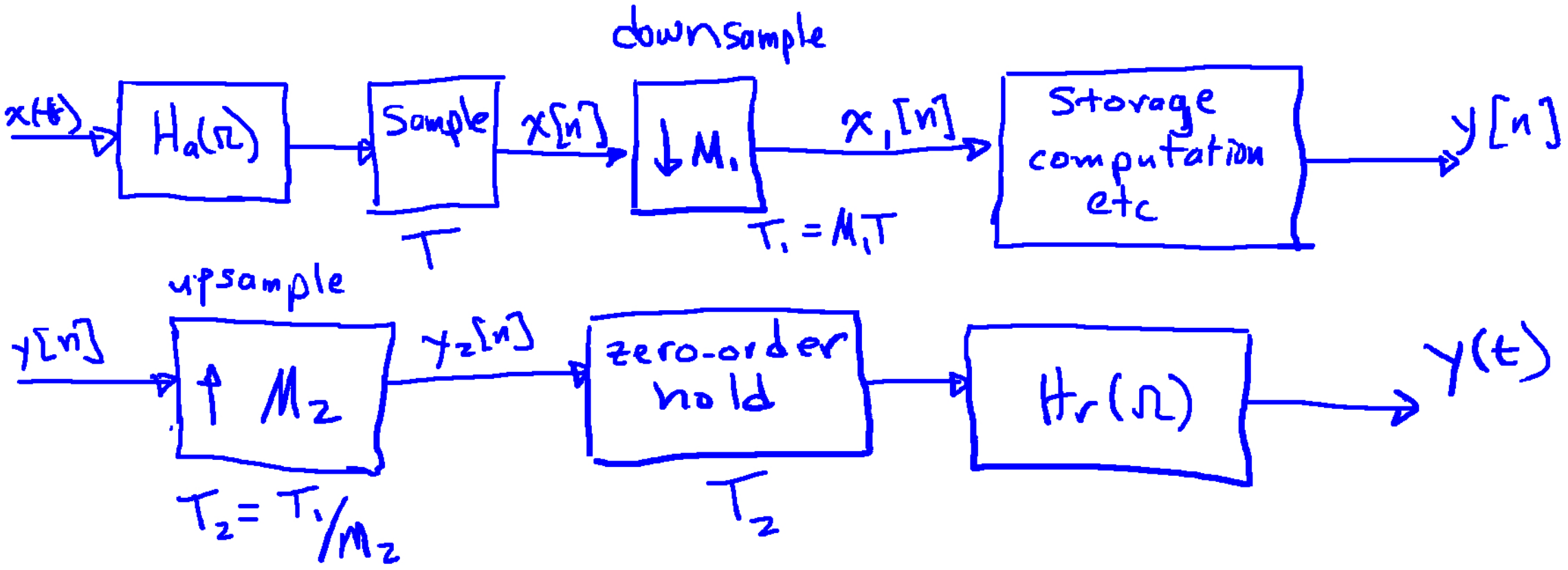
$|H_r(\Omega)|$



$\Rightarrow$ Want $\frac{\pi}{T} >> \Omega_0$ Ω_0: highest freq of interest in $x(t)$

$\Rightarrow$ T small: need fast computation
excessive data storage

Change sampling rate - downsampling, upsampling



low cost – analog filters, storage/computation

price – extra computation for $\downarrow M_1$ and $\uparrow M_2$,
faster A/D and D/A

For more great educational signal-processing content:

AllSignalProcessing.com

- Intuitive, easy-to-follow style
- Organization that saves you time
- Answers to your questions
- Learn at your pace
- Award winning instruction